

Messages Directory App

** Mobile Programming — Exercise 3 **

A React Native message directory app built with **Expo** and **TypeScript**. The app displays a list of message categories. When you tap a category, you see the saved messages for that folder.

Github link: <https://github.com/Roccccc/messages-directory.git>

Features

- Home screen with 5 categories: **Home**, **Love**, **Family**, **Friends**, and **School**
- Category detail screen showing stored messages
- Back navigation from category view to the home screen
- Responsive layout for web, Android, and iOS
- Mock data stored in a separate TypeScript file

Prerequisites

Install these tools before running the project:

Tool	Minimum version	Download
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Node.js	18 or higher	https://nodejs.org
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npm	Comes with Node.js	—
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Expo Go (optional, for phone testing)	Latest	App Store / Google Play
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Optional (for emulators).

- **Android:** Android Studio + Android emulator
- **iOS (Mac only):** Xcode + iOS Simulator

Configuration and Setup

1. Open the project folder

```
``bash
cd "/Users/roc/Desktop/MP/Exercise 3"
``
```

> If you cloned or copied the project to another location, use your own folder path instead.

2. Install dependencies

```
``bash
npm install
``
```

This installs Expo, React Native, React, and TypeScript.

3. Start the development server

```
``bash
npm start
``
```

Expo Dev Tools will open in the terminal. From there you can choose how to run the app.

How to Run the App

Web (Chrome)

```
``bash
npm run web
``
```

Or press `w` after running `npm start`.

The app opens at: ``http://localhost:8081``

Android Emulator

1. Start an Android emulator in Android Studio.
2. Run:

```
```bash
npm run android
```
```

Or press ``a`` after running ``npm start``.

iOS Simulator (Mac only)

```
```bash
npm run ios
```
```

Or press ``i`` after running ``npm start``.

Physical Phone (Expo Go)

1. Install **Expo Go** on your phone.
2. Run ``npm start``.
3. Scan the QR code shown in the terminal with:

- **iPhone:** Camera app
- **Android:** Expo Go app

How to Use the App

1. Open the app — you see the **Messages Directory** home screen.
2. The home screen lists 5 categories with message counts.
3. Tap a category (e.g. **School**) to view its messages.

4. Tap **Back to categories** to return to the home screen.

Screenshots

YOUR INBOX

Messages Directory

Browse important messages grouped by category.

5

Categories

15

Messages

All Categories



Home

3 saved messages



Love

3 saved messages



Family

3 saved messages



Friends

3 saved messages



School

3 saved messages



[◀ Back to categories](#)

HOME



Home

3 messages in this folder

1

MESSAGE

Welcome home! Dinner is ready at 6 PM.

2

MESSAGE

Please take out the trash before you leave.

3

MESSAGE

The Wi-Fi password is on the fridge.

[◀ Back to categories](#)

LOVE



3 messages in this folder

1

MESSAGE

Thinking of you today. Hope your day is going well.

2

MESSAGE

Good morning, sunshine. You make every day brighter.

3

MESSAGE

See you tonight. I picked up your favorite dessert.

◀ Back to categories

FRIENDS



Friends

3 messages in this folder

1

MESSAGE

Game night at my place on Friday. Bring snacks!

2

MESSAGE

Thanks for helping me move last weekend.

3

MESSAGE

Coffee tomorrow at 10 AM? Let me know.

Suggested screenshots

****1. Home screen**** — shows all categories and stats

![Home screen](docs/screenshots/home-screen.png)

****2. Category screen**** — shows messages after tapping a category

![Category screen](docs/screenshots/category-screen.png)

How to capture screenshots

****Web (easiest):****

1. Run `npm run web`

2. Open `http://localhost:8081` in Chrome
3. Press `Cmd + Shift + 4` (Mac) or use your OS screenshot tool
4. Save files into `docs/screenshots/`

****Android Emulator:****

- Click the camera icon in the emulator toolbar

****iOS Simulator:****

- Press `Cmd + S` to save a screenshot

Exact Project Structure

...

Exercise 3/

```
|—— App.tsx           # Root component; switches between Home and Category screens
|—— app.json          # Expo app configuration
|—— index.ts          # App entry point
|—— package.json       # Dependencies and npm scripts
|—— package-lock.json  # Locked dependency versions
|—— tsconfig.json      # TypeScript configuration
|—— .gitignore         # Git ignore rules
|
|—— assets/           # App icons and splash images
|  |—— icon.png
|  |—— favicon.png
|  |—— splash-icon.png
|  |—— android-icon-background.png
|  |—— android-icon-foreground.png
|  |—— android-icon-monochrome.png
|
|—— components/       # Reusable UI components
|  |—— CategoryCard.tsx # Category list item with icon and color
|  |—— MessageItem.tsx  # Single message bubble
|  |—— ScreenContainer.tsx # Responsive centered layout wrapper
|  |—— ScreenHeader.tsx # Header card used on both screens
```

```

|
|—— constants/
|   |—— theme.ts           # Colors and per-category theme (emoji, accent color)
|
|—— data/
|   |—— mockMessages.ts    # Mock message categories and helper functions
|
|—— screens/
|   |—— HomeScreen.tsx     # Lists all message categories
|   |—— CategoryScreen.tsx # Shows messages for one category
|
|—— docs/
|   |—— screenshots/       # README screenshots (add your own PNG files here)
|       |—— home-screen.png
|       |—— category-screen.png
...

```

File Descriptions

| File | Purpose |

|-----|-----|

| `App.tsx` | Main app logic. Uses React state to show either `HomeScreen` or `CategoryScreen`. |

| `data/mockMessages.ts` | Defines `Message` and `MessageCategory` types and stores all mock data. |

| `screens/HomeScreen.tsx` | Renders the category list using `CategoryCard` components. |

| `screens/CategoryScreen.tsx` | Renders messages for the selected category using `MessageItem`. |

| `constants/theme.ts` | Shared colors and category-specific styling (emoji, accent colors). |

| `components/CategoryCard.tsx` | Pressable card for each category on the home screen. |

| `components/MessageItem.tsx` | Styled message row on the category screen. |

Technologies Used

- **React Native** — mobile UI framework
- **Expo** — development and build tooling
- **TypeScript** — typed JavaScript

- **React** — component-based UI

Troubleshooting

| Problem | Solution |
|---------|----------|
|---------|----------|

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|-------|-------|
| ----- | ----- |
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| | |
|---------------------|---|
| `npm install` fails | Make sure Node.js 18+ is installed: `node -v` |
|---------------------|---|

| | |
|--------------------------|---|
| Port 8081 already in use | Stop other Expo servers, or run: `npx expo start --port 8082` |
|--------------------------|---|

| | |
|------------------------|---|
| Web shows a blank page | Stop the server (`Ctrl+C`), run `npm run web` again, hard refresh (`Cmd+Shift+R`) |
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| | |
|----------------------------|---|
| Android emulator not found | Open Android Studio and start an emulator first |
|----------------------------|---|

| | |
|---------------------|---|
| Changes not visible | Reload the app: press `r` in the terminal, or shake the device in Expo Go |
|---------------------|---|

Assignment Checklist

- ☒ React Native app using JS/TS

- ☒ List of message categories on home screen

- ☒ Stored messages shown when a category is clicked

- ☒ README with setup instructions and project structure

- ☐ Add screenshots to `docs/screenshots/`

- ☐ Upload to GitHub

- ☐ Submit ZIP to course management (D2L)

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